

Split Molecular Dynamics

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Goal: Learn MPI communicator concept using *in situ* data analysis of molecular dynamics simulation



MPI_Comm_split()

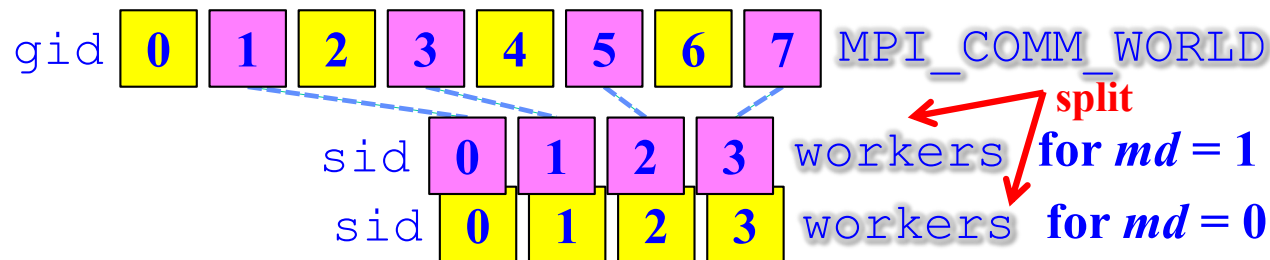
```
MPI_Comm mothercomm, daughtercomm;  
int color, key;  
MPI_Comm_split(mothercomm, color, key, &daughtercomm);
```

- **MPI_Comm_split()** subdivides a communicator, `mothercomm`, into a set of daughter communicators, where processes of the same `color` belong to the same daughter communicator. Processes within each `color` are ranked according to `key`, or if `key` is the same, according to the rank in `mothercomm`. It returns a pointer to a daughter communicator, `daughtercomm`, to which the process belongs.
- **MPI_Comm_split()** is a simpler, higher-level function to construct communicators, instead of using **MPI_Comm_create()** combined with **MPI_Group_excl()** or **MPI_Group_incl()**.

MD & Analysis Communicators

- Split `MPI_COMM_WORLD` into two communicators; one performs molecular dynamics (MD) simulation, whereas the other analyzes simulation data on the fly in background

```
int gid,sid,md;
MPI_Comm workers;
MPI_Comm_rank(MPI_COMM_WORLD,&gid); //Global rank
md = gid%2; // = 1 (MD workers) or 0 (analysis workers)
MPI_Comm_split(MPI_COMM_WORLD,md,0,&workers);
MPI_Comm_rank(workers,&sid); // Rank in workers
```



Run as `mpirun -n 2×nproc`

of processes needed for MD (specified in `pmd.h`)

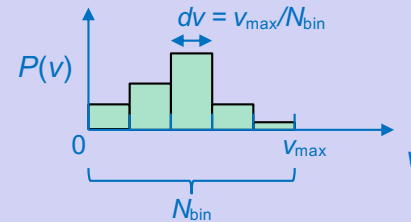
Analysis: Velocity Probability Density

$P(v)$: Probability density function of atom velocity v

```
#define VMAX 5.0 // Maximum velocity value to construct a velocity histogram
#define NBIN 100 // # of bins in the histogram

void calc_pv() {
    double lpv[NBIN],pv[NBIN],dv,v;
    int i;

    dv = VMAX/NBIN; // Bin size
    for (i=0; i<NBIN; i++) lpv[i] = 0.0; // Reset local histogram
    for (i=0; i<n; i++) {
        v = sqrt(pow(rv[i][0],2)+pow(rv[i][1],2)+pow(rv[i][2],2));
        lpv[v/dv < NBIN ? (int)(v/dv) : NBIN-1] += 1.0; // Increment histogram
    }
    MPI_Allreduce(lpv,pv,NBIN,MPI_DOUBLE,MPI_SUM,workers);
    MPI_Allreduce(&n,&nglob,1,MPI_INT,MPI_SUM,workers);
    for (i=0; i<NBIN; i++) pv[i] /= (dv*nglob); // Normalization
    if (sid == 0) {
        for (i=0; i<NBIN; i++) fprintf(fpv,"%le\t%le\n",i*dv,pv[i]);
        fprintf(fpv,"\n");
    }
}
```



$$v = |\vec{v}| = \sqrt{v_x^2 + v_y^2 + v_z^2}$$

$$\int dv p(v) \cong$$

$$dv \sum_{i=0}^{N_{bin}-1} p(v_i) = 1$$

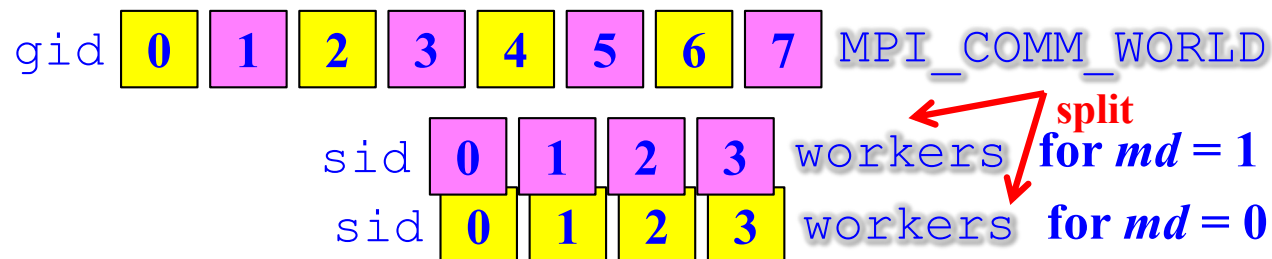
Main Program: Initialization

```
init_params();  
if (md) {  
    set_topology();  
    init_conf();  
    atom_copy();  
    compute_accel();  
}  
else  
    if (sid == 0) fpv = fopen("pv.dat", "w");
```

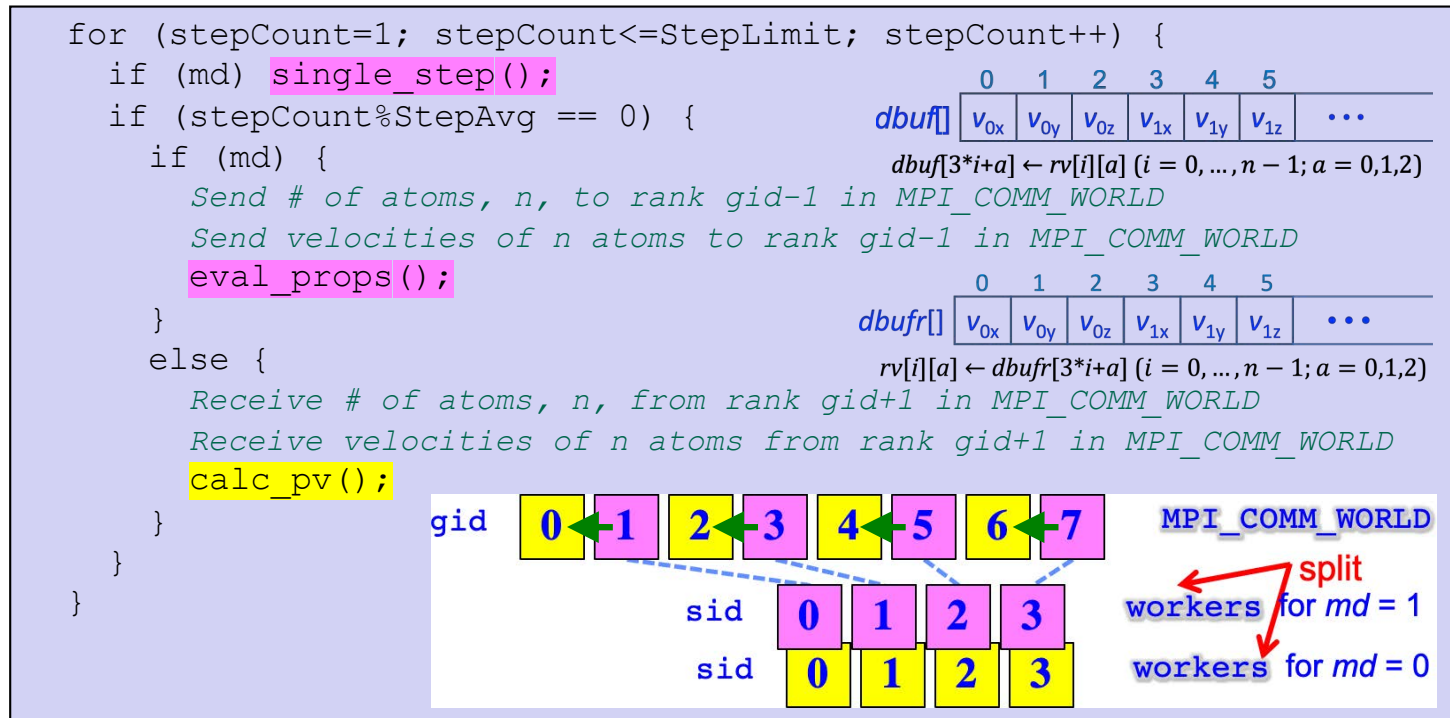
*SPMD (single program multiple data)
Who does what*

Define FILE *fpv;

- All processes read input parameters, `init_params()`. The `nproc` processes of MD workers (`md == 1`) perform MD initialization tasks, whereas only rank 0 among the other `nproc` analysis workers (`md == 0`) opens a file to output the calculated velocity probability density function.



Main Program: Main MD Loop

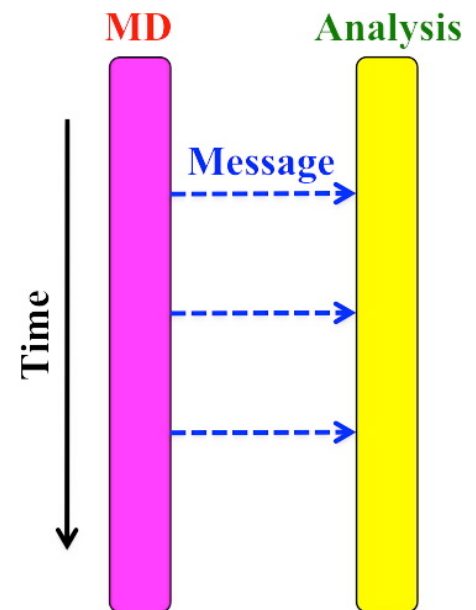


- MD workers perform MD simulation. Every `stepAvg` steps, MD workers send their atom velocities to corresponding analysis workers (*i.e.*, those with the same ranks in respective daughter communicators). Upon receiving the velocities, analysis workers calculate the velocity probability density function.

Main Program: Finalization

```
if (md && sid == 0)
    printf("CPU & COMT = %le %le\n", cpu, comt);
if (!md && sid == 0)
    fclose(fpv);
```

- Rank 0 of MD workers reports the computing & communication times, whereas rank 0 of analysis workers closes the probability density output file
- **Finally:** Change all MPI_COMM_WORLD's in the original MD functions to workers ~ it's only a matter in the small MD world!



Big Picture

- *In situ* handshaking using single-program multiple-data (SPMD) & communicators

Before
main()

```
void calc_pv() {  
    MPI_Allreduce(..., workers);  
}
```

} **Non-MD (analysis) workers
function**
from calc_pv.c

```
int main() {  
    if (md) { SPMD  
        MPI_Send(..., MPI_COMM_WORLD);  
    }  
    // Inter-workers communication  
    // in mother communicator  
}
```

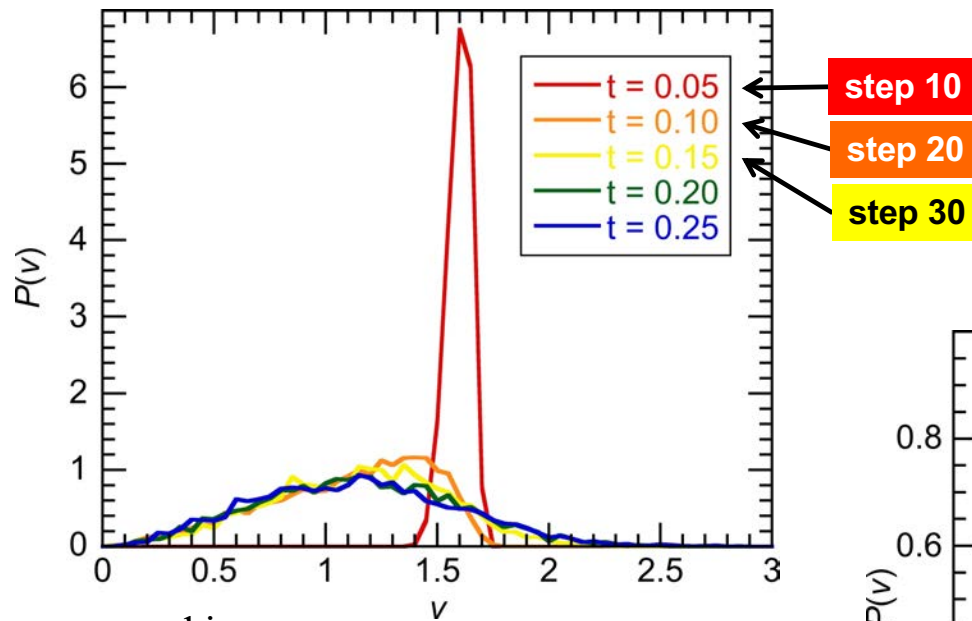
} **MD-analysis hand-shaking**

After
main()

```
void init_params() {...}  
...  
void atom_copy() {  
    MPI_Send(..., workers);  
}  
...  
int bmv() {...}
```

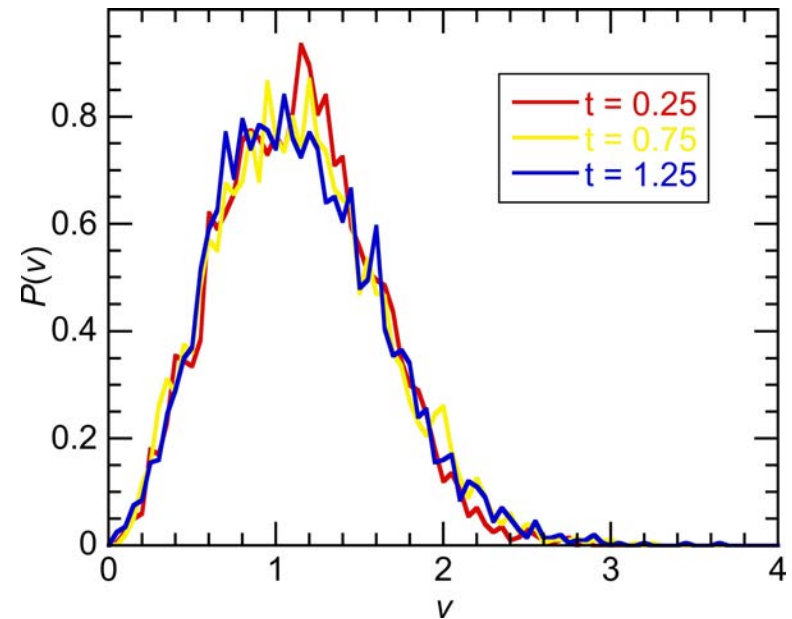
} **MD workers functions**
from pmd_irecv.c

Results



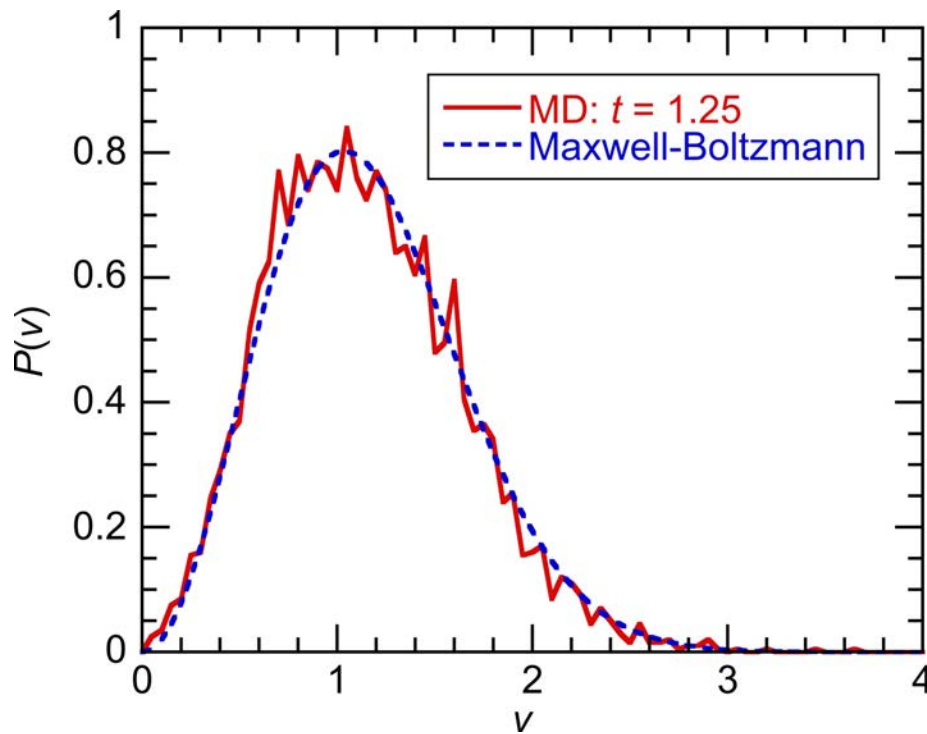
pmd.in

```
5 5 5 // InitUcell[0|1|2]
0.8 // Density
1.0 // InitTemp
0.005 // DeltaT
30 // StepLimit
10 // StepAvg
```

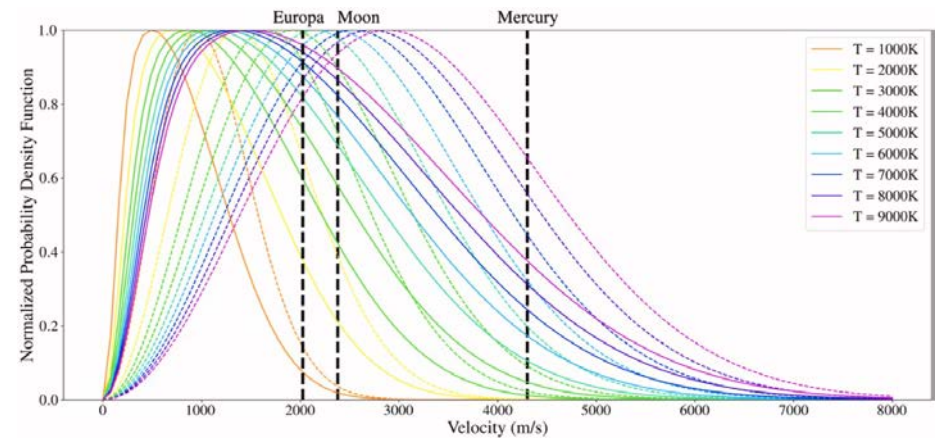


Maxwell-Boltzmann Distribution

$$P_{\text{Maxwell-Boltzmann}}(v) = \frac{4}{\sqrt{\pi}} \left(\frac{m}{2k_{\text{B}}T} \right)^{3/2} v^2 \exp\left(-\frac{mv^2}{2k_{\text{B}}T}\right)$$

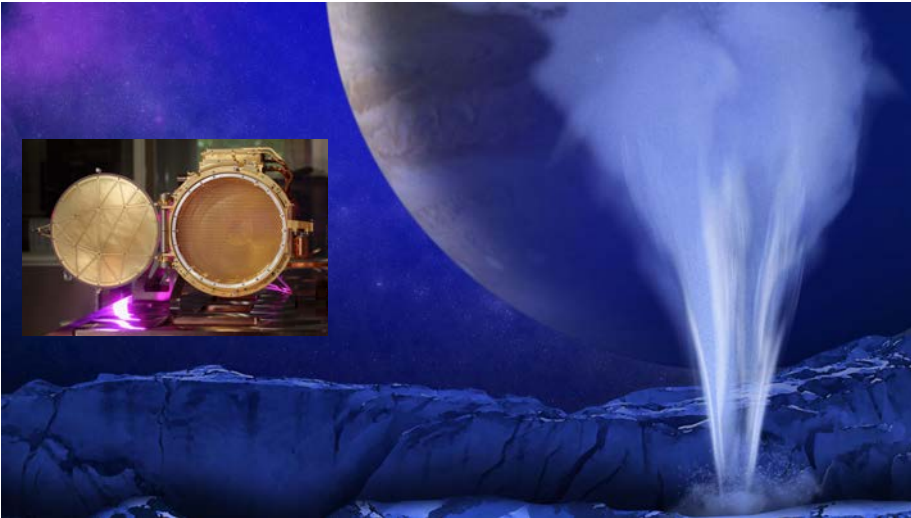


K. Shimamura et al., [Appl. Phys. Lett. 107, 231903 \('15\)](#)



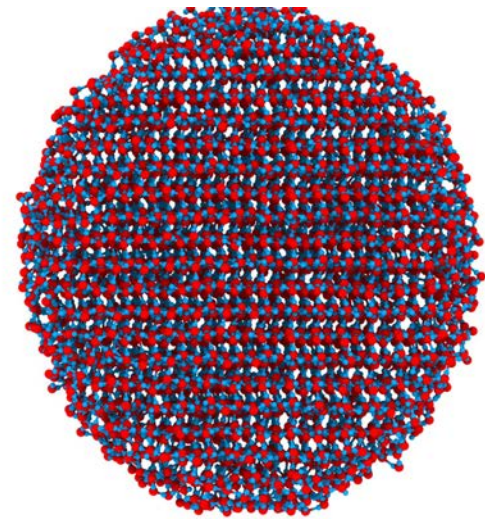
Z. Huang et al., [Icarus 411, 2115947 \('24\)](#)

Maxwell-Boltzmann Distribution in Europa



<https://www.youtube.com/watch?v=4QJS9LcB66g>

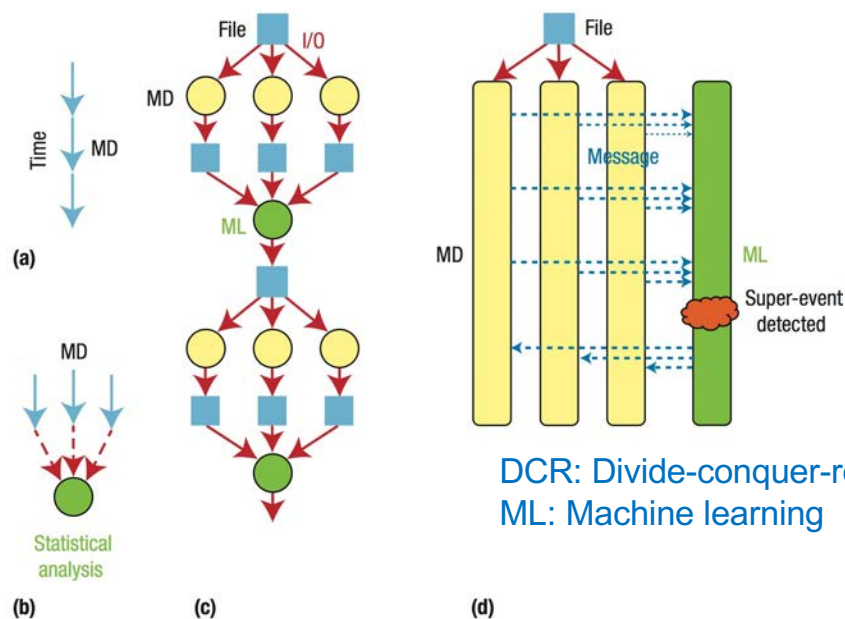
**Europa Clipper mission will arrive in 2030
to observe water geyser**



Z. Huang *et al.*, [*Icarus* 411, 2115947 \('24\)](#)

In Situ Data Analysis

Use communicators to add data analytics & extra logic to parallel simulations



N. A. Romero *et al.*,
[*IEEE Computer* **48**\(11\), 33 \('15\)](#)

FIGURE 2. DCR in time. (a) Molecular dynamics (MD) simulations have sequential time dependence. (b) Parallel replica dynamics (PRD) predicts long-time behavior through statistical analysis of multiple parallel MD trajectories. (c) Conventional file-based and (d) new in situ PRD simulations. ML represents machine-learning tasks.

See also T. Do *et al.*, [A lightweight method for evaluating *in situ* workflow efficiency](#), *J. Comput. Sci.* **48**, 101259 ('21)